

Lab08: Unsupervised Learning Mini-Project

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Background

The purpose of this mini-project is to carry out a full analysis using the unsupervised learning methods introduced in our previous class.

The dataset comes from the Wisconsin Breast Cancer Diagnostic dataset, originally reported by K. P. Benne and O. L. Mangasarian in their work “Robust Linear Programming Discrimination of Two Linearly Inseparable Sets.”

The values in this dataset represent features of cell nuclei observed in digitized images obtained from a fine needle aspiration (FNA) of a breast mass.

Data import

```
wisc.df <- read.csv("C:/Users/chenr/Downloads/WisconsinCancer.csv", row.names = 1)
head(wisc.df)
```

```
##      diagnosis radius_mean texture_mean perimeter_mean area_mean
## 842302         M      17.99       10.38         122.80      1001.0
## 842517         M      20.57       17.77         132.90      1326.0
## 84300903        M      19.69       21.25         130.00      1203.0
## 84348301         M      11.42       20.38          77.58       386.1
## 84358402         M      20.29       14.34         135.10      1297.0
## 843786         M      12.45       15.70          82.57       477.1
##      smoothness_mean compactness_mean concavity_mean concave.points_mean
## 842302         0.11840         0.27760         0.3001         0.14710
## 842517         0.08474         0.07864         0.0869         0.07017
## 84300903        0.10960         0.15990         0.1974         0.12790
## 84348301        0.14250         0.28390         0.2414         0.10520
## 84358402        0.10030         0.13280         0.1980         0.10430
## 843786         0.12780         0.17000         0.1578         0.08089
##      symmetry_mean fractal_dimension_mean radius_se texture_se perimeter_se
## 842302         0.2419         0.07871         1.0950         0.9053         8.589
## 842517         0.1812         0.05667         0.5435         0.7339         3.398
## 84300903        0.2069         0.05999         0.7456         0.7869         4.585
## 84348301        0.2597         0.09744         0.4956         1.1560         3.445
## 84358402        0.1809         0.05883         0.7572         0.7813         5.438
## 843786         0.2087         0.07613         0.3345         0.8902         2.217
##      area_se smoothness_se compactness_se concavity_se concave.points_se
## 842302      153.40      0.006399      0.04904      0.05373      0.01587
## 842517       74.08      0.005225      0.01308      0.01860      0.01340
## 84300903      94.03      0.006150      0.04006      0.03832      0.02058
```

```
## 84348301 27.23 0.009110 0.07458 0.05661 0.01867
## 84358402 94.44 0.011490 0.02461 0.05688 0.01885
## 843786 27.19 0.007510 0.03345 0.03672 0.01137
## symmetry_se fractal_dimension_se radius_worst texture_worst
## 842302 0.03003 0.006193 25.38 17.33
## 842517 0.01389 0.003532 24.99 23.41
## 84300903 0.02250 0.004571 23.57 25.53
## 84348301 0.05963 0.009208 14.91 26.50
## 84358402 0.01756 0.005115 22.54 16.67
## 843786 0.02165 0.005082 15.47 23.75
## perimeter_worst area_worst smoothness_worst compactness_worst
## 842302 184.60 2019.0 0.1622 0.6656
## 842517 158.80 1956.0 0.1238 0.1866
## 84300903 152.50 1709.0 0.1444 0.4245
## 84348301 98.87 567.7 0.2098 0.8663
## 84358402 152.20 1575.0 0.1374 0.2050
## 843786 103.40 741.6 0.1791 0.5249
## concavity_worst concave.points_worst symmetry_worst
## 842302 0.7119 0.2654 0.4601
## 842517 0.2416 0.1860 0.2750
## 84300903 0.4504 0.2430 0.3613
## 84348301 0.6869 0.2575 0.6638
## 84358402 0.4000 0.1625 0.2364
## 843786 0.5355 0.1741 0.3985
## fractal_dimension_worst
## 842302 0.11890
## 842517 0.08902
## 84300903 0.08758
## 84348301 0.17300
## 84358402 0.07678
## 843786 0.12440
```

Remove the diagnosis from data for subsequent analysis

```
wisc.data <- wisc.df[,-1]
dim(wisc.data)
```

```
## [1] 569 30
```

The first column diagnosis is the expert opinion on the sample (i.e. patient FNA).

```
head(wisc.df$diagnosis)
```

```
## [1] "M" "M" "M" "M" "M" "M"
```

```
head(wisc.df[,-1])
```

```
## radius_mean texture_mean perimeter_mean area_mean smoothness_mean
## 842302 17.99 10.38 122.80 1001.0 0.11840
## 842517 20.57 17.77 132.90 1326.0 0.08474
## 84300903 19.69 21.25 130.00 1203.0 0.10960
## 84348301 11.42 20.38 77.58 386.1 0.14250
## 84358402 20.29 14.34 135.10 1297.0 0.10030
## 843786 12.45 15.70 82.57 477.1 0.12780
## compactness_mean concavity_mean concave.points_mean symmetry_mean
## 842302 0.27760 0.3001 0.14710 0.2419
## 842517 0.07864 0.0869 0.07017 0.1812
```

```
## 84300903      0.15990      0.1974      0.12790      0.2069
## 84348301      0.28390      0.2414      0.10520      0.2597
## 84358402      0.13280      0.1980      0.10430      0.1809
## 843786        0.17000      0.1578      0.08089      0.2087
##      fractal_dimension_mean radius_se texture_se perimeter_se area_se
## 842302          0.07871      1.0950      0.9053          8.589 153.40
## 842517          0.05667      0.5435      0.7339          3.398  74.08
## 84300903        0.05999      0.7456      0.7869          4.585  94.03
## 84348301        0.09744      0.4956      1.1560          3.445  27.23
## 84358402        0.05883      0.7572      0.7813          5.438  94.44
## 843786          0.07613      0.3345      0.8902          2.217  27.19
##      smoothness_se compactness_se concavity_se concave.points_se
## 842302          0.006399      0.04904      0.05373          0.01587
## 842517          0.005225      0.01308      0.01860          0.01340
## 84300903        0.006150      0.04006      0.03832          0.02058
## 84348301        0.009110      0.07458      0.05661          0.01867
## 84358402        0.011490      0.02461      0.05688          0.01885
## 843786          0.007510      0.03345      0.03672          0.01137
##      symmetry_se fractal_dimension_se radius_worst texture_worst
## 842302          0.03003          0.006193          25.38          17.33
## 842517          0.01389          0.003532          24.99          23.41
## 84300903        0.02250          0.004571          23.57          25.53
## 84348301        0.05963          0.009208          14.91          26.50
## 84358402        0.01756          0.005115          22.54          16.67
## 843786          0.02165          0.005082          15.47          23.75
##      perimeter_worst area_worst smoothness_worst compactness_worst
## 842302          184.60      2019.0          0.1622          0.6656
## 842517          158.80      1956.0          0.1238          0.1866
## 84300903        152.50      1709.0          0.1444          0.4245
## 84348301          98.87       567.7          0.2098          0.8663
## 84358402        152.20      1575.0          0.1374          0.2050
## 843786          103.40       741.6          0.1791          0.5249
##      concavity_worst concave.points_worst symmetry_worst
## 842302          0.7119          0.2654          0.4601
## 842517          0.2416          0.1860          0.2750
## 84300903        0.4504          0.2430          0.3613
## 84348301        0.6869          0.2575          0.6638
## 84358402        0.4000          0.1625          0.2364
## 843786          0.5355          0.1741          0.3985
##      fractal_dimension_worst
## 842302          0.11890
## 842517          0.08902
## 84300903        0.08758
## 84348301        0.17300
## 84358402        0.07678
## 843786          0.12440
```

Now store the diagnosis as a vector for use later when we compare our results to those from experts in the field.

```
diagnosis <- factor(wisc.df$diagnosis)
```

Q1. How many observations are in the dataset?

569 observations in the dataset.

Q2. How many of the observations have a malignant diagnosis?

```
table(wisc.df$diagnosis)
```

```
##
##      B      M
## 357 212
```

Q3. How many variables/features in the data are suffixed with `_mean`?

```
#colnames(wisc.data)
length(grep("_mean", colnames(wisc.data)))
```

```
## [1] 10
```

Principal Component Analysis

The `prcomp()` function for PCA uses `scale = FALSE` by default. In most cases, we set this to `TRUE` so that the analysis isn't dominated by variables with larger means or standard deviations simply due to differences in measurement units or scales.

```
wisc.pr <- prcomp(wisc.data, scale=TRUE)
summary(wisc.pr)
```

```
## Importance of components:
##
##              PC1      PC2      PC3      PC4      PC5      PC6      PC7
## Standard deviation  3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
## Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
## Cumulative Proportion 0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
##
##              PC8      PC9      PC10     PC11     PC12     PC13     PC14
## Standard deviation  0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
## Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
## Cumulative Proportion 0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
##
##              PC15     PC16     PC17     PC18     PC19     PC20     PC21
## Standard deviation  0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
## Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
## Cumulative Proportion 0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
##
##              PC22     PC23     PC24     PC25     PC26     PC27     PC28
## Standard deviation  0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
## Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
## Cumulative Proportion 0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
##
##              PC29     PC30
## Standard deviation  0.02736 0.01153
## Proportion of Variance 0.00002 0.00000
## Cumulative Proportion 1.00000 1.00000
```

Q4. From your results, what proportion of the original variance is captured by the first principal components (PC1)?

PC1 Proportion of Variance = 0.4427, which is 44.27%, So 44.27% of the variance is captured by PC1.

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

After PC3 0.72636 (≥ 0.70), the first point that meets the target, so 3 PCs.

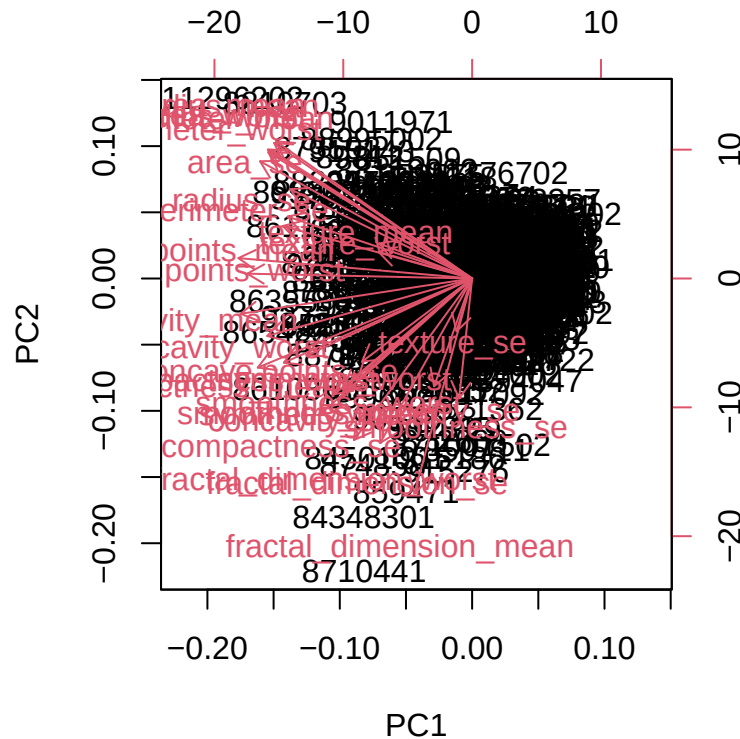
Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

After PC7 0.91010 (≥ 0.90), the first point that meets the target, so 7 PCs.

Interpreting PCA results

Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

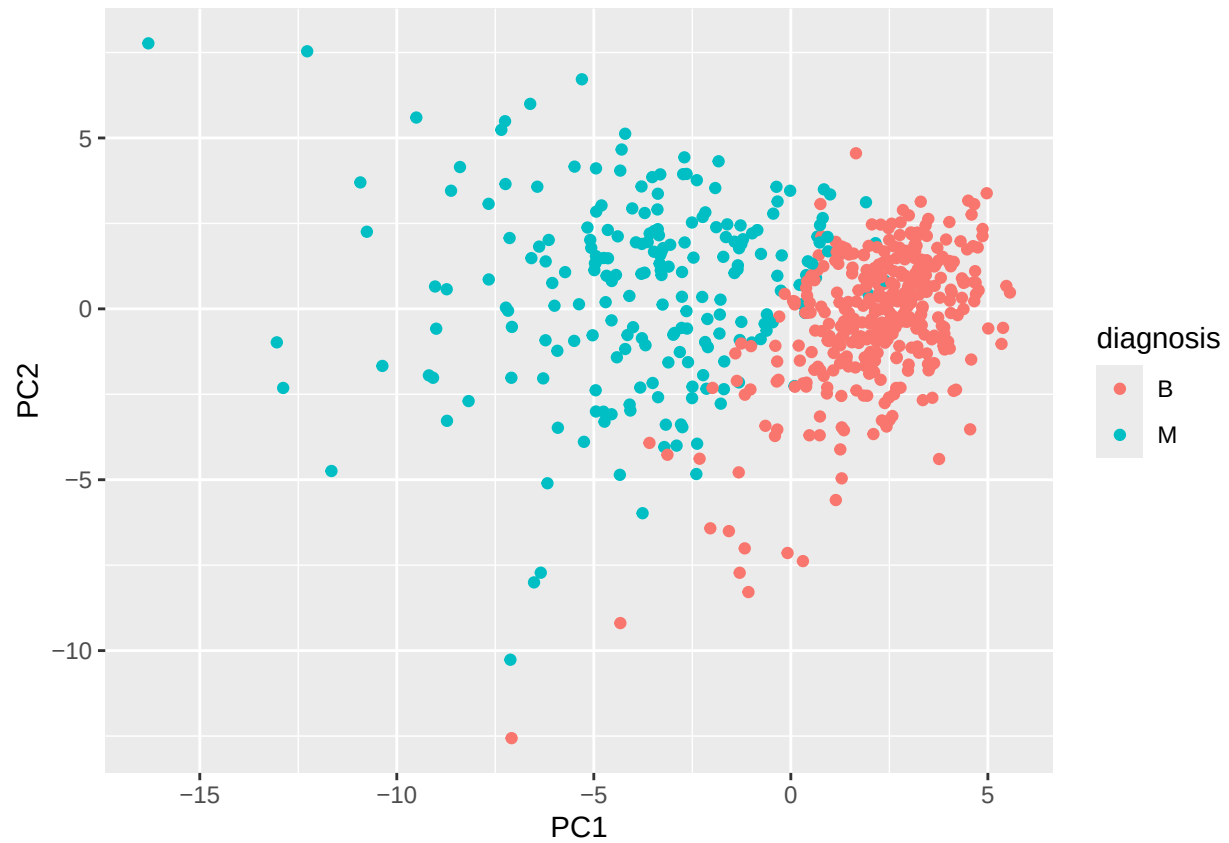
```
biplot(wisc.pr)
```



The biplot is difficult to interpret because the rownames clutter the plot, making trends and groupings hard to see in this high-dimensional dataset.

```
library(ggplot2)

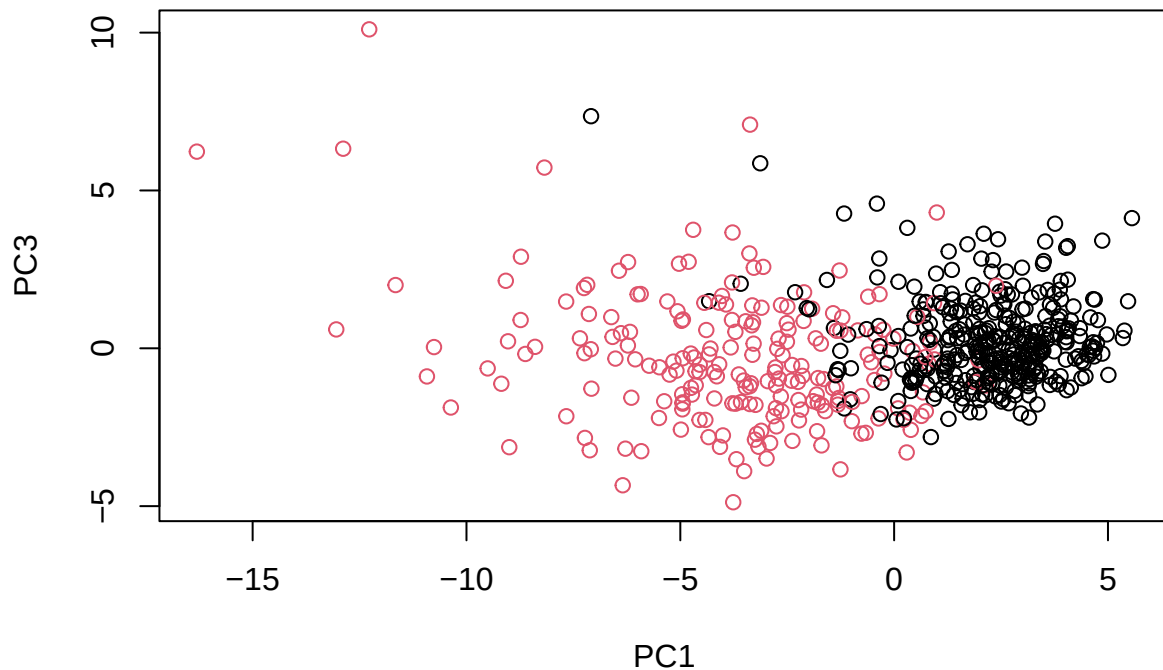
ggplot(wisc.pr$x) +
  aes(PC1, PC2, col=diagnosis) +
  geom_point()
```



Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

Repeat for components 1 and 3

```
plot(wisc.pr$x[, c(1, 3)], col = diagnosis, xlab = "PC1", ylab = "PC3")
```



The PC1 vs. PC3 plot shows less clear separation between malignant and benign samples than the PC1 vs. PC2 plot. This is because the third principal component captures less variance than the second, resulting in less distinct groupings.

Variance explained

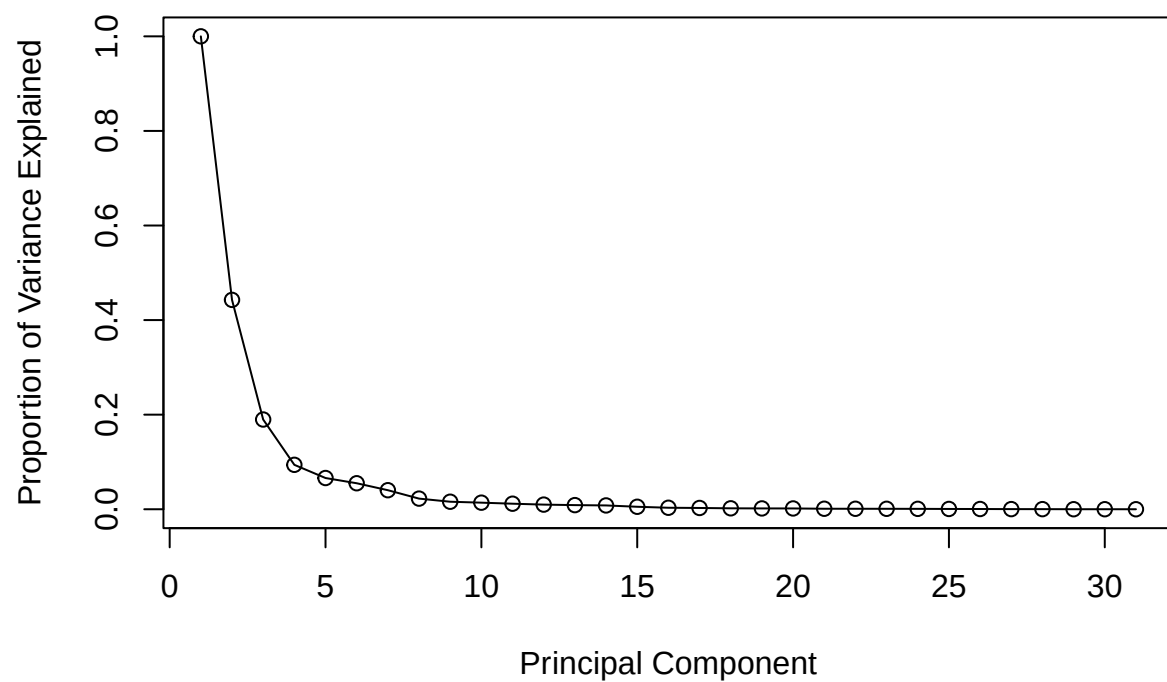
Variance of each PC

```
# Calculate variance of each component
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

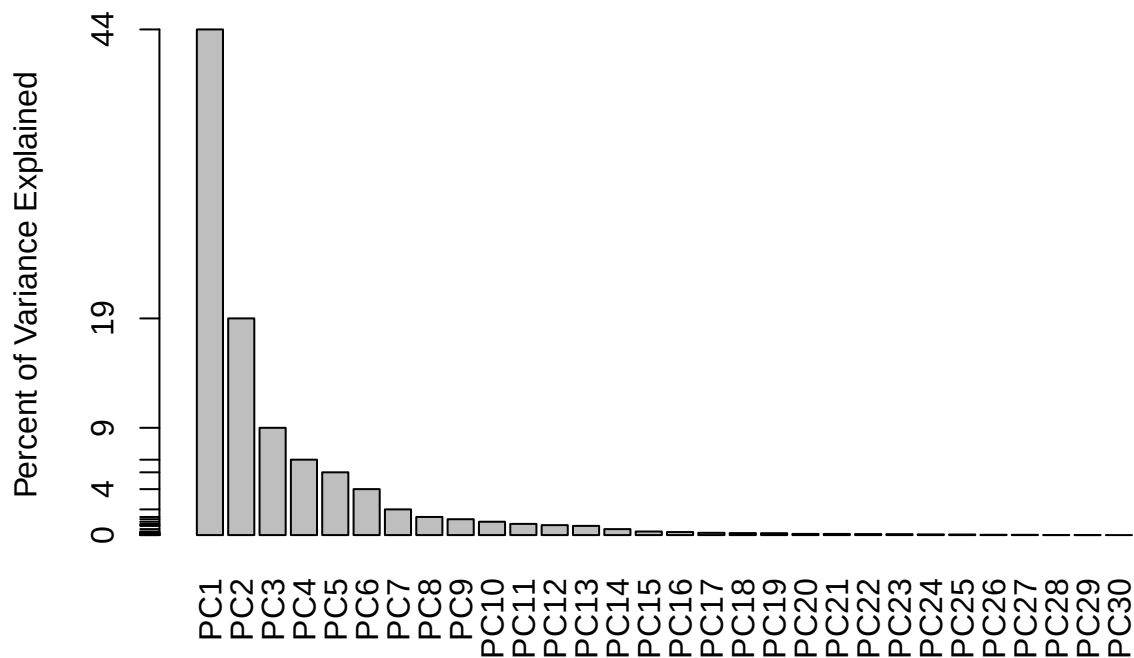
```
## [1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

Proportion of variance explained

```
# Variance explained by each principal component: pve
pve <- pr.var / sum(pr.var)
# Plot variance explained for each principal component
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```



```
# Alternative scree plot of the same data, note data driven y-axis  
barplot(pve, ylab = "Percent of Variance Explained",  
        names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)  
axis(2, at=pve, labels=round(pve,2)*100 )
```

Communicating PCA results

Q9. For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
wisc.pr$rotation["concave.points_mean", "PC1"]
```

```
## [1] -0.2608538
```

We need 5 PCs to capture more than 80% variance.

Q10. What is the minimum number of principal components required to explain 80% of the variance of the data?

```
summary(wisc.pr)
```

```
## Importance of components:
##          PC1      PC2      PC3      PC4      PC5      PC6      PC7
## Standard deviation  3.6444  2.3857  1.67867  1.40735  1.28403  1.09880  0.82172
## Proportion of Variance 0.4427  0.1897  0.09393  0.06602  0.05496  0.04025  0.02251
## Cumulative Proportion 0.4427  0.6324  0.72636  0.79239  0.84734  0.88759  0.91010
##          PC8      PC9      PC10     PC11     PC12     PC13     PC14
## Standard deviation  0.69037  0.6457  0.59219  0.5421  0.51104  0.49128  0.39624
## Proportion of Variance 0.01589  0.0139  0.01169  0.0098  0.00871  0.00805  0.00523
## Cumulative Proportion 0.92598  0.9399  0.95157  0.9614  0.97007  0.97812  0.98335
##          PC15     PC16     PC17     PC18     PC19     PC20     PC21
## Standard deviation  0.30681  0.28260  0.24372  0.22939  0.22244  0.17652  0.1731
```

```
## Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
## Cumulative Proportion 0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
##                      PC22  PC23  PC24  PC25  PC26  PC27  PC28
## Standard deviation    0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
## Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
## Cumulative Proportion 0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
##                      PC29  PC30
## Standard deviation    0.02736 0.01153
## Proportion of Variance 0.00002 0.00000
## Cumulative Proportion 1.00000 1.00000
```

Hierarchical clustering

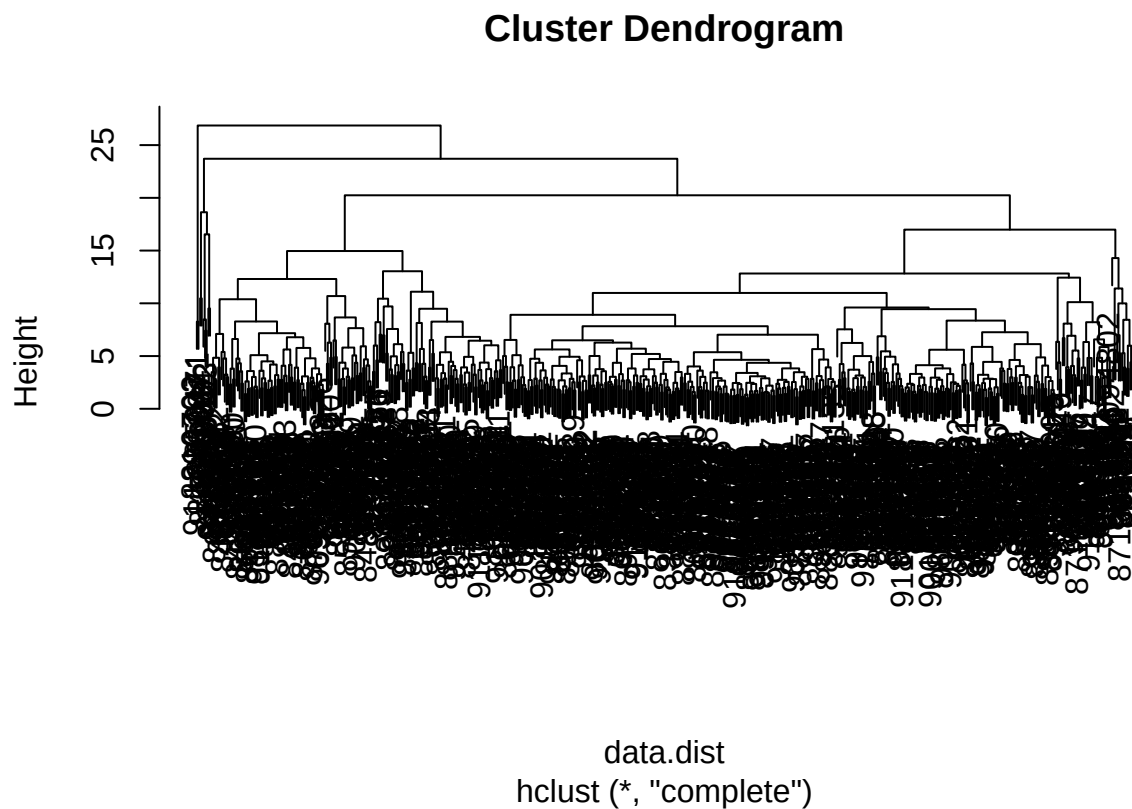
Just clustering the original data is not very informative or helpful.

Q11. Using the `plot()` and `abline()` functions, what is the height at which the clustering model has 4 clusters?

```
data.scaled <- scale(wisc.data)
data.dist <- dist(data.scaled)
wisc.hclust <- hclust(data.dist)
```

View the clustering dendrogram result

```
plot(wisc.hclust)
```



```
wisc.hclust.clusters <- cutree(wisc.hclust, k=4)
table(wisc.hclust.clusters)
```

```
## wisc.hclust.clusters
##   1   2   3   4
## 177   7 383   2

table(wisc.hclust.clusters, diagnosis)
```

```
##           diagnosis
## wisc.hclust.clusters  B  M
##           1  12 165
##           2   2   5
##           3 343  40
##           4   0   2
```

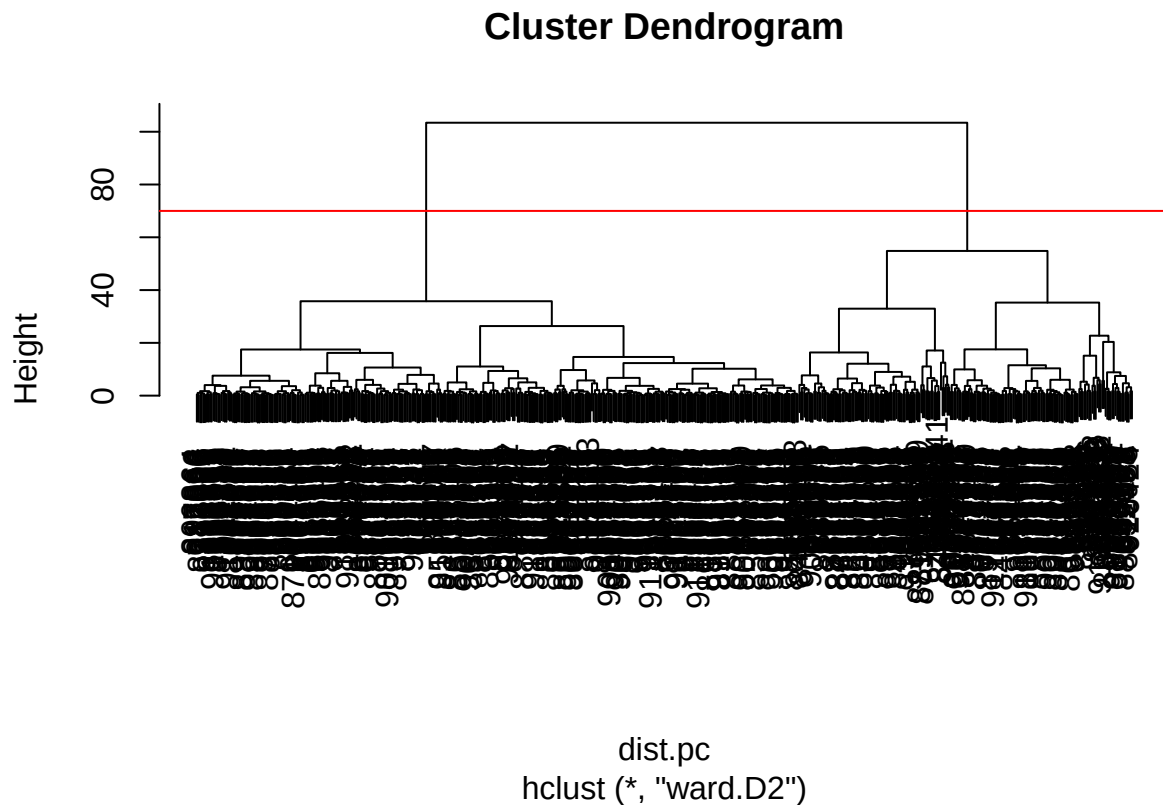
Combing methods (PCA and Clustering)

Clustering the original data was not very effective, whereas the PCA results appeared more promising. Here, we combine these approaches by performing clustering on the PCA output—in other words, clustering in principal component (PC) space.

```
## Take the first 3 PCs
dist.pc <- dist(wisc.pr$x[,1:3])
wisc.pr.hclust <- hclust(dist.pc, method="ward.D2")
```

View the tree...

```
plot(wisc.pr.hclust)
abline(h=70, col="red")
```



To get our clustering membership vector (i.e. our main clustering result) we “cut” the tree at a desired height or to yield a desired number of “k” groups.

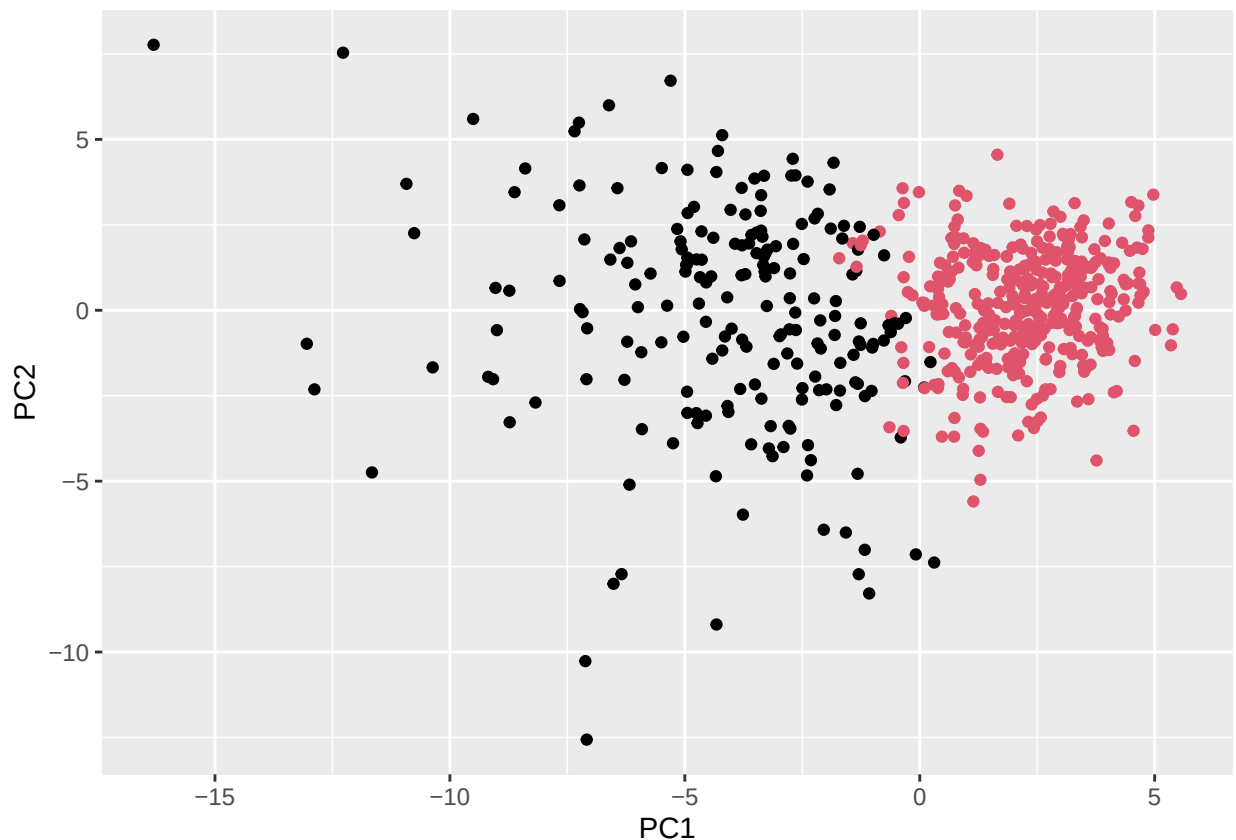
```
grps <- cutree(wisc.pr.hclust, h=70)
table(grps)
```

```
## grps
##    1    2
## 203 366
```

```
table(grps, diagnosis)
```

```
##      diagnosis
## grps    B    M
##    1   24 179
##    2  333  33
```

```
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col=grps)
```



Q12. Which method gives your favorite results for the same data.dist dataset? Explain your reasoning. or How well does the newly created model with four clusters separate out the two diagnoses?

Hierarchical clustering on the PCA-transformed data gives the best results because PCA reduces noise and highlights the main structure. The four-cluster model partially separates malignant and benign cases but still shows some overlap.

Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output

of each model (`wisc.hclust.clusters` and `wisc.pr.hclust.clusters`).

Hierarchical clustering on the original data performs poorly, as clusters contain a mix of malignant and benign samples. For example, cluster 1 is mostly malignant while cluster 3 is mostly benign, but neither achieves clean separation.

Q16. Which of these new patients should we prioritize for follow up based on your results?

Patients assigned to clusters enriched with malignant cases (such as cluster 1) should be prioritized for follow-up. These clusters contain the highest proportion of malignant diagnoses, indicating higher risk.